SUJAL KUMAR

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PROFILE

B.Tech Mathematics & Computing student at NIT Jalandhar with hands-on experience building software applications, applied AI/ML systems, and backend services. Open-source contributor interested in software engineering, algorithms, and production-grade applications.

EDUCATION

National Institute of Technology (NIT), Jalandhar **2025–2029**
B.Tech, Mathematics & Computing Jalandhar, India

- **9.02 CGPA** — among the top 10% of the batch; relevant coursework: Python and Machine Learning.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Dart, Java

Computer Science: Data Structures & Algorithms, OOP, Problem Solving, STL

Backend & Data: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, relational data modelling

AI / Information Retrieval: RAG, LLMs, embeddings, semantic chunking, vector search, hybrid retrieval, TensorFlow

Cloud & Developer Tools: Azure OpenAI, Azure AI Search, Docker, Git, GitHub, AWS (S3, EC2), Jupyter

EXPERIENCE

Software Developer — Campus Node **2026–Present**
NIT Jalandhar Jalandhar, India

- Developing a campus event-management application to centralize event discovery, organizer workflows, and student participation.
- Designing application workflows and features for managing campus events, organizers, and event information.

Open Source Contributor — GirlScript Summer of Code **May–July 2026**
GirlScript Foundation Remote / Jalandhar, India

- Collaborated with mentors and developers through Git/GitHub; resolved issues and implemented feature enhancements within contribution and review workflows.

PROJECTS

Financial Intelligence & RAG Engine **2026**
Python, FastAPI, Azure OpenAI, Azure AI Search, PostgreSQL, Docker GitHub

- Built an end-to-end ingestion pipeline for 10-K/10-Q filings: PyMuPDF4LLM converts complex PDFs into structured Markdown while retaining financial tables, headings, and disclosure context.
- Implemented context-aware semantic chunking and embedding-based indexing in Azure AI Search; hybrid keyword/vector retrieval supplies relevant filing evidence for downstream analysis.
- Used GPT-4o to extract quantitative KPIs (revenue, net income, operating cash flow, assets and liabilities) and qualitative growth drivers/risk factors; normalized results and persisted structured records in PostgreSQL.
- Developed FastAPI endpoints and a financial analytics dashboard for KPI inspection, fiscal-year scoping, cross-company comparison, and conversational answers grounded in retrieved filing excerpts.
- **Technology distinction:** Azure AI Search handles retrieval, GPT-4o performs extraction and synthesis, and PostgreSQL stores structured metrics—separating unstructured evidence retrieval from reliable analytical queries.

Exchange Matching Engine **In Progress**
C++ Systems Programming

- Developing a systems-programming project focused on the core exchange workflow: order intake, order-book management, matching, and trade execution.
- Emphasizing predictable execution paths, efficient memory use, and data-structure choices suited to latency-sensitive workloads rather than general-purpose application abstractions.
- **Technology distinction:** C++ provides fine-grained control over object lifetime, memory layout, and allocation; the project explores these systems-level trade-offs in an exchange-style workload.
- **Current status:** in progress; performance figures and completed feature claims are intentionally omitted until benchmarked and verified.

Sanjeevani — Herbal Wellness Application **Jan–Feb 2026**

Flutter, Dart, Firebase GitHub

- Built a cross-platform wellness application with symptom-based recommendations, Firebase Authentication, user profiles, and cart-management workflows.
- Implemented a Material Design 3 interface for Android, iOS, and Web, using Firebase services for authentication and application data workflows.

CERTIFICATIONS

AI/ML for Geodata Analytics | Indian Institute of Remote Sensing (IIRS), ISRO **Sep 2026**

- Completed training focused on applying artificial intelligence and machine learning techniques to geospatial data and analytics.

Cyber Job Simulation | Deloitte **Nov 2026**

- Completed a simulated cybersecurity job experience, gaining exposure to practical, role-oriented cyber security work.

ACHIEVEMENTS & ACTIVITIES

- **Solved 250+ DSA questions** on LeetCode.
- **Among the top 10 teams** in BIS Innovation Hackathon.
- **Organizing team member for Utkansh**, the annual cultural festival of NIT Jalandhar; supported event coordination and management.